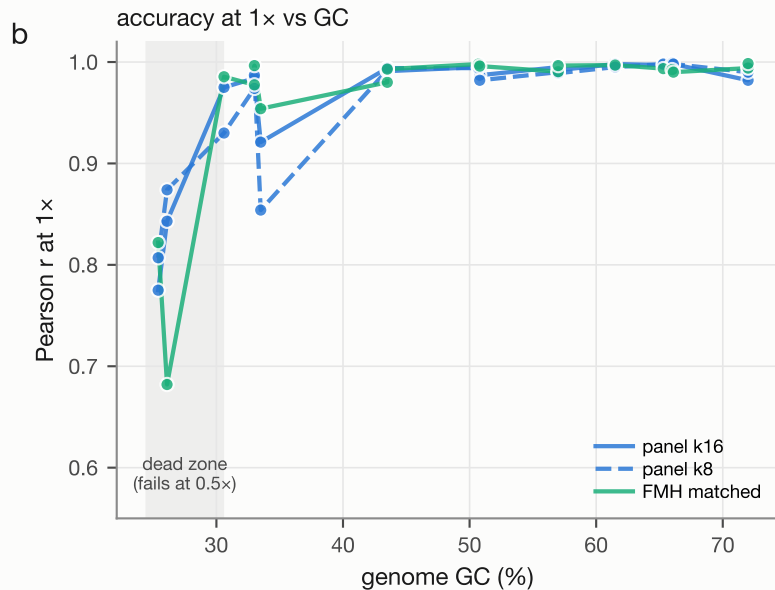
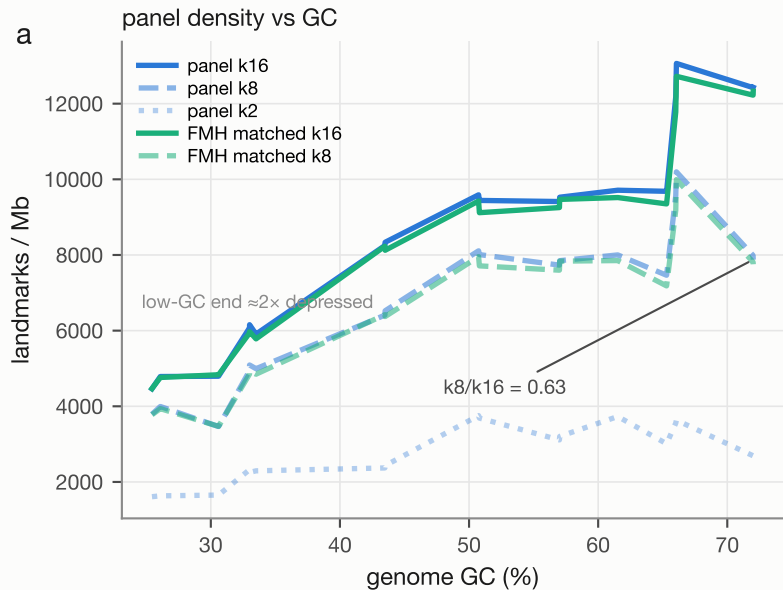




Eighteen real genomes spanning GC 25.4–72.0%, simulated reads, planted $\log_2$ PTR 0.5–2.0 ($n = 8$ cells per point). a, Enzyme-panel landmark density against GC: k16 rises monotonically ($\approx 4.5\text{k/Mb}$ at 26% to $\approx 12.4\text{k/Mb}$ at 72%) and only the low-GC end is depressed ($\approx 2\times$ below mid-GC); density-matched FracMinHash tracks the panel within a few percent at every GC. The k8/k16 ratio thins at the GC extremes (0.63 at 72%) — the extra eight enzymes buy high-GC density, not accuracy. b, Pearson r at 1 $\times$: k8 matches k16 at every GC despite carrying only 63% of its landmarks at 72% GC (A6 holds across GC), and matched FracMinHash ties the panel. Shaded band: the instrument boundary — at GC $\leq 30\%$ together with 0.5 $\times$ depth every landmark mode fails ($r < 0.7$) — so the claimable range is GC $\geq 30\%$ at $\geq 1\times$ coverage.